

2106.07890 — formula report

319 inline formulas, 51 display equations, 0 tables, 6 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2106.07890_E00004	007	■ 0.255 ● C +9	$\begin{cases} * \rightarrow * & \text{when } d \text{ contains } c, \text{ red, and blue} \\ \emptyset \rightarrow * & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does contain blue} \\ \emptyset \rightarrow \emptyset & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does not contain blue.} \end{cases}$	$\begin{cases} * \rightarrow * & \text{when } d \text{ contains } c, \text{ red, and blue} \\ \emptyset \rightarrow * & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does contain blue} \\ \emptyset \rightarrow \emptyset & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does not contain blue.} \end{cases}$	$\begin{cases} * \rightarrow * & \text{when } d \text{ contains } c, \text{ red, and blue} \\ \emptyset \rightarrow * & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does contain blue} \\ \emptyset \rightarrow \emptyset & \text{when } d \text{ does not contain both } c \text{ and red, and } d \text{ does not contain blue.} \end{cases}$
2106.07890_E00012	011	■ 0.403 ● W +2	$\begin{aligned} & \mathcal{L}(\text{red,red firetruck}) = 0.02 \\ & \mathcal{L}(\text{red,red idea}) = 10^{-5} \\ & \mathcal{L}(\text{red, blue sky}) = 0 \end{aligned}$	$\mathcal{L}(\text{red,red firetruck}) = 0.02$ $\mathcal{L}(\text{red,red idea}) = 10^{-5}$ $\mathcal{L}(\text{red, blue sky}) = 0$	$\mathcal{L}(\text{red,red firetruck}) = 0.02$ $\mathcal{L}(\text{red,red idea}) = 10^{-5}$ $\mathcal{L}(\text{red,blue sky}) = 0.$
2106.07890_E00030	018	■ 0.508 ● C +4	$\begin{aligned} & \lim_{W} F(c) = \min \left\{ \frac{\mathcal{L}(c,d)f(c)}{w_1}, \frac{\mathcal{L}(c,d)g(c)}{w_2}, \mathcal{L}(c,d) \right\} \\ & \leq \min \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2}, 1 \right\} \\ & \leq \min \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2}, 1 \right\} \\ & = \lim_{W} F(d) \end{aligned}$	$\mathcal{L}(c,d) \lim_{W} F(c) = \min \left\{ \frac{\mathcal{L}(c,d)f(c)}{w_1}, \frac{\mathcal{L}(c,d)g(c)}{w_2}, \mathcal{L}(c,d) \right\}$ $\leq \min \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2}, 1 \right\}$ $\leq \min \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2}, 1 \right\}$ $= \lim_{W} F(d)$	$\mathcal{L}(c,d) \lim_{W} F(c) = \min \left\{ \frac{\mathcal{L}(c,d)f(c)}{w_1}, \frac{\mathcal{L}(c,d)g(c)}{w_2}, \mathcal{L}(c,d) \right\}$ $\leq \min \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2}, 1 \right\}$ $\leq \min \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2}, 1 \right\}$ $= \lim_{W} F(d),$
2106.07890_E00024 (16)	017	■ 0.538 ● K +0	$\mathcal{E}(\operatorname{colim}_W F, -) \cong \mathcal{V}^{\mathcal{J}^{\operatorname{op}}}(W, \mathcal{E}(F, -))$	$\mathcal{E}(\operatorname{colim}_W F, -) \cong \mathcal{V}^{\mathcal{J}^{\operatorname{op}}}(W, \mathcal{E}(F, -))$	$\mathcal{E}(\operatorname{colim}_W F, -) \cong \mathcal{V}^{\mathcal{J}^{\operatorname{op}}}(W, \mathcal{E}(F, -))$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2106.07890_E00034	019	■0.573 ●W +2	$\begin{aligned} \widehat{\mathcal{L}}(h, f \times g) &= \inf_{c \in \mathcal{L}} \{[hc, (f \times g)c]\} \\ &= \inf_{c \in \mathcal{L}} \{[hc, \min\{fc, gc\}]\} \\ &= \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{hc}, \frac{gc}{hc}, 1 \right\} \\ &= \min_{\{ \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{hc}, 1 \right\}, \inf_{c \in \mathcal{L}} \left\{ \frac{gc}{hc}, 1 \right\} \}} \left\{ \frac{fc}{hc}, \frac{gc}{hc}, 1 \right\} \\ &= \widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g) \end{aligned}$	$\begin{aligned} \widehat{\mathcal{L}}(h, f \times g) &= \inf_{c \in \mathcal{L}} \{[hc, (f \times g)c]\} \\ &= \inf_{c \in \mathcal{L}} \{[hc, \min\{fc, gc\}]\} \\ &= \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{hc}, \frac{gc}{hc}, 1 \right\} \\ &= \min_{\{ \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{hc}, 1 \right\}, \inf_{c \in \mathcal{L}} \left\{ \frac{gc}{hc}, 1 \right\} \}} \left\{ \frac{fc}{hc}, \frac{gc}{hc}, 1 \right\} \\ &= \widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g) \end{aligned}$	
2106.07890_E00044	021	■0.657 ●W +1	$\begin{aligned} \widehat{\mathcal{L}}(h^x, h^y)(c) &= \inf_{d \in \mathcal{L}} \{[(h^c \times h^x)(d), h^y(d)]\} \\ &= \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\pi(d c) \times \pi(d x)}, 1 \right\} \\ &= \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\min\{\pi(d c), \pi(d x)\}}, 1 \right\}. \end{aligned}$	$\begin{aligned} [h^x, h^y](c) &= \widehat{\mathcal{L}}(h^c \times h^x, h^y) \\ &= \inf_{d \in \mathcal{L}} \{[(h^c \times h^x)(d), h^y(d)]\} \\ &= \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\pi(d c) \times \pi(d x)}, 1 \right\} \\ &= \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\min\{\pi(d c), \pi(d x)\}}, 1 \right\}. \end{aligned}$	$\begin{aligned} [h^x, h^y](c) &= \widehat{\mathcal{L}}(h^c \times h^x, h^y) \\ &= \inf_{d \in \mathcal{L}} \{[(h^c \times h^x)(d), h^y(d)]\} \\ &= \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\pi(d c) \times \pi(d x)}, 1 \right\} \\ &= \inf_{d \in \mathcal{L}} \left\{ \frac{\pi(d y)}{\min\{\pi(d c), \pi(d x)\}}, 1 \right\}. \end{aligned}$
2106.07890_E00003 (2)	007	■0.667 ●N +0	$\left((h^c \times h^{\text{red}})(d) \rightarrow h^{\text{blue}}(d) \right)_{d \in \text{ob}(\mathcal{C})}$	$\left\{ \left(h^c \times h^{\text{red}} \right) (d) \rightarrow h^{\text{blue}}(d) \right\}_{d \in \text{ob}(\mathcal{C})}$	$\left\{ (h^c \times h^{\text{red}})(d) \rightarrow h^{\text{blue}}(d) \right\}_{d \in \text{ob}(\mathcal{C})}$
2106.07890_E00020 (12)	014	■0.776 ●N +1	$c \mapsto h^x(c) := \begin{cases} \pi(c x) & \text{if } x \leq c \\ 0 & \text{otherwise} \end{cases}$	$c \mapsto h^x(c) := \begin{cases} \pi(c x) & \text{if } x \leq c \\ 0 & \text{otherwise} \end{cases}$	$c \mapsto h^x(c) := \begin{cases} \pi(c x) & \text{if } x \leq c \\ 0 & \text{otherwise.} \end{cases}$
2106.07890_E00032 (19)	018	■0.783 ●C +10	$\left((w_1, h^x) \times (w_2, h^y) \right)(c) = \begin{cases} \min \left\{ \frac{\pi(c x)}{w_1}, \frac{\pi(c y)}{w_2}, 1 \right\} & \text{if } x \leq c \text{ and } y \leq c \\ 0 & \text{otherwise} \end{cases}$	$\left((w_1, h^x) \times (w_2, h^y) \right)(c) = \begin{cases} \min \left\{ \frac{\pi(c x)}{w_1}, \frac{\pi(c y)}{w_2}, 1 \right\} & \text{if } x \leq c \text{ and } y \leq c \\ 0 & \text{otherwise} \end{cases}$	$\left((w_1, h^x) \times (w_2, h^y) \right)(c) = \begin{cases} \min \left\{ \frac{\pi(c x)}{w_1}, \frac{\pi(c y)}{w_2}, 1 \right\} & \text{if } x \leq c \text{ and } y \leq c \\ 0 & \text{otherwise.} \end{cases}$
2106.07890_E00005	007	■0.797 ●C +5	$\left(h^x, h^y \right)(c) = \begin{cases} * & \text{if every text that contains both } c \text{ and } x \text{ also contains } y \\ \emptyset & \text{otherwise} \end{cases}$	$[h^x, h^y](c) = \begin{cases} * & \text{if every text that contains both } c \text{ and } x \text{ also contains } y \\ \emptyset & \text{otherwise} \end{cases}$	$[h^x, h^y](c) = \begin{cases} * & \text{if every text that contains both } c \text{ and } x \text{ also contains } y \\ \emptyset & \text{otherwise.} \end{cases}$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured					

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2106.07890_E00088 (6)	010	0.890 C +3	$[a, b] := \begin{cases} b/a & \text{if } b < a \\ 1 & \text{otherwise} \end{cases}$	$[a, b] := \begin{cases} b/a & \text{if } b < a \\ 1 & \text{otherwise} \end{cases}.$	$[a, b] := \begin{cases} b/a & \text{if } b < a, \\ 1 & \text{otherwise.} \end{cases}$
2106.07890_E00040	020	0.955 W +1	$\begin{aligned} & \operatorname{colim}_W F(c) = \max \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2} \right\} \\ & = \operatorname{colim}_W F(d), \end{aligned}$	$\mathcal{L}(c, d) \operatorname{colim}_W F(c) = \max \{ \mathcal{L}(c, d) w_1 f(c), \mathcal{L}(c, d) w_2 g(c) \}$ $\leq \max \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2} \right\}$ $= \operatorname{colim}_W F(d),$	$\mathcal{L}(c, d) \operatorname{colim}_W F(c) = \max \{ \mathcal{L}(c, d) w_1 f(c), \mathcal{L}(c, d) w_2 g(c) \}$ $\leq \max \left\{ \frac{f(d)}{w_1}, \frac{g(d)}{w_2} \right\}$ $= \operatorname{colim}_W F(d),$
2106.07890_E00048	023	0.967 W +2	$\begin{aligned} & \left(\left(w_1, f \right) \times \left(w_2, g \right) \right) (c) = -\ln \left(\left(\left(w'_1, f' \right) \times \left(w'_2, g' \right) \right) (c) \right) \\ & = -\ln \left(\min \left\{ \frac{f'c}{w'_1}, \frac{g'c}{w'_2}, 1 \right\} \right) \\ & = \max \left\{ -\ln \left(\frac{f'c}{w'_1} \right), -\ln \left(\frac{g'c}{w'_2} \right), 0 \right\} \\ & = \max \{ -\ln(f'c) + \ln(w'_1), -\ln(g'c) + \ln(w'_2), 0 \} \\ & = \max \{ fc - w_1, gc - w_2, 0 \}. \end{aligned}$	$\begin{aligned} & ((w_1, f) \times (w_2, g))(c) = -\ln(((w'_1, f') \times (w'_2, g'))(c)) \\ & = -\ln\left(\min\left\{\frac{f'c}{w'_1}, \frac{g'c}{w'_2}, 1\right\}\right) \\ & = \max\left\{-\ln\left(\frac{f'c}{w'_1}\right), -\ln\left(\frac{g'c}{w'_2}\right), 0\right\} \\ & = \max\{-\ln(f'c) + \ln(w'_1), -\ln(g'c) + \ln(w'_2), 0\} \\ & = \max\{fc - w_1, gc - w_2, 0\} \end{aligned}$	$\begin{aligned} ((w_1, f) \times (w_2, g))(c) &= -\ln(((w'_1, f') \times (w'_2, g'))(c)) \\ &= -\ln\left(\min\left\{\frac{f'c}{w'_1}, \frac{g'c}{w'_2}, 1\right\}\right) \\ &= \max\left\{-\ln\left(\frac{f'c}{w'_1}\right), -\ln\left(\frac{g'c}{w'_2}\right), 0\right\} \\ &= \max\{-\ln(f'c) + \ln(w'_1), -\ln(g'c) + \ln(w'_2), 0\} \\ &= \max\{fc - w_1, gc - w_2, 0\}. \end{aligned}$
2106.07890_E00029	017	0.990 W +1	$\begin{aligned} & [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F)) = \min \left\{ \left[w_1, \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{Zc}, 1 \right\} \right], \left[w_2, \inf_{c \in \mathcal{L}} \left\{ \frac{gc}{Zc}, 1 \right\} \right] \right\} \\ & = \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1 \right\}. \end{aligned}$	$[0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F)) = \min \left\{ \left[w_1, \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{Zc}, 1 \right\} \right], \left[w_2, \inf_{c \in \mathcal{L}} \left\{ \frac{gc}{Zc}, 1 \right\} \right] \right\}$ $= \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1 \right\}.$	$[0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F)) = \min \left\{ \left[w_1, \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{Zc}, 1 \right\} \right], \left[w_2, \inf_{c \in \mathcal{L}} \left\{ \frac{gc}{Zc}, 1 \right\} \right] \right\}$ $= \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1 \right\}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2106.07890_E00037	019	■0.996 ●W +0	$\begin{aligned} & \widehat{\operatorname{colim}_W Fi, Z} = \inf_{c \in \mathcal{L}} \{ \max \{ w_1 f c, w_2 g c \}, Z c \} \\ & = \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{\max \{ w_1 f c, w_2 g c \}}, 1 \right\} \\ & = \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{w_1 f c}, \frac{Z c}{w_2 g c} \right\}. \end{aligned}$	$\widehat{\mathcal{L}}(\operatorname{colim}_W Fi, Z) = \inf_{c \in \mathcal{L}} \{ [\max \{ w_1 f c, w_2 g c \}, Z c] \}$ $= \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{\max \{ w_1 f c, w_2 g c \}}, 1 \right\}$ $= \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{w_1 f c}, \frac{Z c}{w_2 g c} \right\}.$	$\widehat{\mathcal{L}}(\operatorname{colim}_W Fi, Z) = \inf_{c \in \mathcal{L}} \{ [\max \{ w_1 f c, w_2 g c \}, Z c] \}$ $= \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{\max \{ w_1 f c, w_2 g c \}}, 1 \right\}$ $= \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{w_1 f c}, \frac{Z c}{w_2 g c} \right\}.$
2106.07890_E00039	020	■0.997 ●C +3	$\begin{aligned} & \{ [0, 1]^{\widehat{\mathcal{L}}(F, Z)} \} = \min \left\{ \left[w_1, \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{f c}, 1 \right\} \right], \left[w_2, \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{g c}, 1 \right\} \right] \right\} \\ & = \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{w_1 f c}, \frac{Z c}{w_2 g c}, 1 \right\}. \end{aligned}$	$[0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z)) = \min \left\{ \left[w_1, \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{f c}, 1 \right\} \right], \left[w_2, \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{g c}, 1 \right\} \right] \right\}$ $= \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{w_1 f c}, \frac{Z c}{w_2 g c}, 1 \right\}.$	$[0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z)) = \min \left\{ \left[w_1, \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{f c}, 1 \right\} \right], \left[w_2, \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{g c}, 1 \right\} \right] \right\}$ $= \inf_{c \in \mathcal{L}} \left\{ \frac{Z c}{w_1 f c}, \frac{Z c}{w_2 g c}, 1 \right\}.$
2106.07890_E00035	019	■0.999 ●K +0	$\left(w_1, f \right) \sqcup \left(w_2, g \right) := \operatorname{colim}_W F.$	$(w_1, f) \sqcup (w_2, g) := \operatorname{colim}_W F.$	$(w_1, f) \sqcup (w_2, g) := \operatorname{colim}_W F.$
2106.07890_E00001	007	■1.000 ●N +1	$\left[h^{\text{red}}, h^{\text{blue}} \right] : \mathcal{C} \rightarrow \mathbf{Set}$	$\left[h^{\text{red}}, h^{\text{blue}} \right] : \mathcal{C} \rightarrow \mathbf{Set}$	$[h^{\text{red}}, h^{\text{blue}}] : \mathcal{C} \rightarrow \mathbf{Set}$
2106.07890_E00010	010	■1.000 ●N -1	$a \times b := \min \{ a, b \}$ $a \sqcup b := \max \{ a, b \}.$	$a \times b := \min \{ a, b \}$ $a \sqcup b := \max \{ a, b \}.$	$a \times b := \min \{ a, b \}$ $a \sqcup b := \max \{ a, b \}.$
2106.07890_E00002 (1)	007	■1.000 ●W -1	$\widehat{\left(h^{\text{red}}, h^{\text{blue}} \right)} = \widehat{\left(h^c, [h^{\text{red}}, h^{\text{blue}}] \right)} = \widehat{\left(h^c \times h^{\text{red}}, h^{\text{blue}} \right)}.$	$\left[h^{\text{red}}, h^{\text{blue}} \right] (c) = \widehat{\mathcal{C}} \left(h^c, [h^{\text{red}}, h^{\text{blue}}] \right) = \widehat{\mathcal{C}} \left(h^c \times h^{\text{red}}, h^{\text{blue}} \right).$	$[h^{\text{red}}, h^{\text{blue}}](c) = \widehat{\mathcal{C}}(h^c, [h^{\text{red}}, h^{\text{blue}}]) = \widehat{\mathcal{C}}(h^c \times h^{\text{red}}, h^{\text{blue}}).$
2106.07890_E00006 (4)	009	■1.000 ●N +0	$1 \leq \mathcal{C}(x, x)$ $\mathcal{C}(y, z) \otimes \mathcal{C}(x, y) \leq \mathcal{C}(x, z)$	$1 \leq \mathcal{C}(x, x)$ $\mathcal{C}(y, z) \otimes \mathcal{C}(x, y) \leq \mathcal{C}(x, z)$	$1 \leq \mathcal{C}(x, x)$ $\mathcal{C}(y, z) \otimes \mathcal{C}(x, y) \leq \mathcal{C}(x, z)$
2106.07890_E00007 (5)	009	■1.000 ●N -2	$x \otimes y \leq z \text{ if and only if } x \leq [y, z]$	$x \otimes y \leq z \text{ if and only if } x \leq [y, z]$	$x \otimes y \leq z \text{ if and only if } x \leq [y, z]$
2106.07890_E00009 (7)	010	■1.000 ●N -2	$ab \leq c \text{ if and only if } a \leq [b, c]$	$ab \leq c \text{ if and only if } a \leq [b, c]$	$ab \leq c \text{ if and only if } a \leq [b, c]$
2106.07890_E00011	011	■1.000 ●N +0	$\mathcal{L}(x, y) := \pi(y \mid x),$	$\mathcal{L}(x, y) := \pi(y \mid x),$	$\mathcal{L}(x, y) := \pi(y x),$
2106.07890_E00013 (8)	011	■1.000 ●K +0	$\pi(z \mid y) \pi(y \mid x) = \pi(z \mid x)$	$\pi(z \mid y) \pi(y \mid x) = \pi(z \mid x)$	$\pi(z y) \pi(y x) = \pi(z x)$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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2106.07890_ E00014 (9)	012	■ 1.000 ● K +0	$\mathrm{C}(x, y) \leq \mathrm{D}(x, y)$	$\mathcal{C}(x, y) \leq \mathcal{D}(x, y)$	$\mathcal{C}(x, y) \leq \mathcal{D}(x, y)$
2106.07890_ E00015 (10)	012	■ 1.000 ● N +0	$\mathrm{D}^{\mathrm{C}}(f, g) := \int_{c \in \mathcal{C}} \mathrm{D}(fc, gc)$	$\mathcal{D}^{\mathcal{C}}(f, g) := \int_{c \in \mathcal{C}} \mathcal{D}(fc, gc)$	$\mathcal{D}^{\mathcal{C}}(f, g) := \int_{c \in \mathcal{C}} \mathcal{D}(fc, gc)$
2106.07890_ E00016 (11)	013	■ 1.000 ● N +0	$\widehat{\mathrm{C}}(f, g) = \inf_{c \in \mathcal{C}} \{\mathrm{C}(fc, gc)\}$	$\widehat{\mathcal{C}}(f, g) = \inf_{c \in \mathcal{C}} \{\mathcal{C}(fc, gc)\}$	$\widehat{\mathcal{C}}(f, g) = \inf_{c \in \mathcal{C}} \{\mathcal{C}(fc, gc)\}$
2106.07890_E00017	013	■ 1.000 ● K +0	$h^x := \mathrm{C}(x, -)$	$h^x := \mathcal{C}(x, -)$	$h^x := \mathcal{C}(x, -)$
2106.07890_E00018	013	■ 1.000 ● K +0	$\mathrm{C}(c, d) \leq [\mathrm{C}(c), \mathrm{C}(d)]$	$\mathcal{C}(c, d) \leq [h^x(c), h^x(d)]$	$\mathcal{C}(c, d) \leq [h^x(c), h^x(d)]$
2106.07890_E00019	013	■ 1.000 ● N +0	$\widehat{\mathrm{C}}(h^x, f) \leq [\mathrm{C}(h^x(x), f)] = [1, f] = f$	$\widehat{\mathcal{C}}(h^x, f) \leq [h^x(x), f] = [1, f] = f$	$\widehat{\mathcal{C}}(h^x, f) \leq [h^x(x), f] = [1, f] = f$
2106.07890_E00021	015	■ 1.000 ● N +0	$x \xrightarrow{a} y \quad h^y \xrightarrow{a} h^x$	$x \xrightarrow{a} y \quad h^y \xrightarrow{a} h^x$	$x \xrightarrow{a} y \quad h^y \xrightarrow{a} h^x$
2106.07890_ E00022 (13)	016	■ 1.000 ● N +1	$\mathrm{C}(-, \lim F) \cong \mathrm{Set}^{\mathrm{J}}(*, \mathrm{C}(-, F))$	$\mathcal{C}(-, \lim F) \cong \mathrm{Set}^{\mathrm{J}}(*, \mathcal{C}(-, F))$	$\mathcal{C}(-, \lim F) \cong \mathrm{Set}^{\mathrm{J}}(*, \mathcal{C}(-, F))$
2106.07890_ E00023 (15)	016	■ 1.000 ● N +1	$\mathrm{E} \left(-, \lim_{\mathcal{W}} F \right) \cong \mathrm{V}^{\mathcal{J}}(\mathcal{W}, \mathrm{E}(-, F))$	$\mathcal{E} \left(-, \lim_{\mathcal{W}} F \right) \cong \mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(-, F))$	$\mathcal{E} \left(-, \lim_{\mathcal{W}} F \right) \cong \mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(-, F))$
2106.07890_E00025	017	■ 1.000 ● N +0	$\left(w_1, f \right) \times \left(w_2, g \right) := \lim_{\mathcal{W}} F$	$(w_1, f) \times (w_2, g) := \lim_{\mathcal{W}} F$	$(w_1, f) \times (w_2, g) := \lim_{\mathcal{W}} F$
2106.07890_ E00026 (17)	017	■ 1.000 ● N +0	$\widehat{\mathrm{L}} \left(Z, \lim_{\mathcal{W}} F \right) \cong [0, 1]^{\mathcal{J}}(W, \widehat{\mathrm{L}}(Z, F))$	$\widehat{\mathcal{L}} \left(Z, \lim_{\mathcal{W}} F \right) \cong [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F))$	$\widehat{\mathcal{L}}(Z, \lim_{\mathcal{W}} F) \cong [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F))$
2106.07890_E00027	017	■ 1.000 ● N +0	$\inf_{c \in \mathcal{L}} \left\{ \left[\mathrm{C}(c, \min \left\{ \frac{fc}{w_1}, \frac{gc}{w_2}, 1 \right\} \right] \right\} = \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{w_1 \mathrm{C}(c)}, \frac{gc}{w_2 \mathrm{C}(c)}, 1 \right\}$	$\inf_{c \in \mathcal{L}} \left\{ \left[Zc, \min \left\{ \frac{fc}{w_1}, \frac{gc}{w_2}, 1 \right\} \right] \right\} = \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1 \right\}$	$\inf_{c \in \mathcal{L}} \left\{ \left[Zc, \min \left\{ \frac{fc}{w_1}, \frac{gc}{w_2}, 1 \right\} \right] \right\} = \inf_{c \in \mathcal{L}} \left\{ \frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1 \right\}$
2106.07890_E00028	017	■ 1.000 ● N +0	$\widehat{\mathrm{L}}(Z, Fi) = \inf_{c \in \mathcal{L}} \{ \mathrm{C}(Zc, Fic) \} = \inf_{c \in \mathcal{L}} \left\{ \frac{Fic}{Zc}, 1 \right\}$	$\widehat{\mathcal{L}}(Z, Fi) = \inf_{c \in \mathcal{L}} \{ [Zc, Fic] \} = \inf_{c \in \mathcal{L}} \left\{ \frac{Fic}{Zc}, 1 \right\}$	$\widehat{\mathcal{L}}(Z, Fi) = \inf_{c \in \mathcal{L}} \{ [Zc, Fic] \} = \inf_{c \in \mathcal{L}} \left\{ \frac{Fic}{Zc}, 1 \right\}$
2106.07890_ E00031 (18)	018	■ 1.000 ● N +1	$\left((w_1, h^x) \times (w_2, h^y) \right) (c) = \min \left\{ \frac{h^x(c)}{w_1}, \frac{h^y(c)}{w_2}, 1 \right\}$	$((w_1, h^x) \times (w_2, h^y))(c) = \min \left\{ \frac{h^x(c)}{w_1}, \frac{h^y(c)}{w_2}, 1 \right\}$	$((w_1, h^x) \times (w_2, h^y))(c) = \min \left\{ \frac{h^x(c)}{w_1}, \frac{h^y(c)}{w_2}, 1 \right\}$
2106.07890_E00033	018	■ 1.000 ● N +0	$\widehat{\mathrm{L}}(h, f \times g) = \widehat{\mathrm{L}}(h, f) \times \widehat{\mathrm{L}}(h, g)$	$\widehat{\mathcal{L}}(h, f \times g) = \widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g)$	$\widehat{\mathcal{L}}(h, f \times g) = \widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g)$
2106.07890_ E00036 (20)	019	■ 1.000 ● K +0	$\widehat{\mathrm{L}}(\mathrm{colim}_{\mathcal{W}} F, Z) = [0, 1]^{\mathcal{J}}(W, \widehat{\mathrm{L}}(F, Z))$	$\widehat{\mathcal{L}}(\mathrm{colim}_{\mathcal{W}} F, Z) = [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z))$	$\widehat{\mathcal{L}}(\mathrm{colim}_{\mathcal{W}} F, Z) = [0, 1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z))$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2106.07890_E00038	019	■ 1.000 ● W +1	$\widehat{\mathcal{L}}(F i, Z)=\inf _{c \in \mathcal{L}}\left\{F i c, Z c\right\}=\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{F i c}, 1\right\} .$	$\widehat{\mathcal{L}}(F i, Z)=\inf _{c \in \mathcal{L}}\left\{F i c, Z c\right\}=\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{F i c}, 1\right\} .$	$\widehat{\mathcal{L}}(F i, Z)=\inf _{c \in \mathcal{L}}\left\{F i c, Z c\right\}=\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{F i c}, 1\right\} .$
2106.07890_E00041	020	■ 1.000 ● N +0	$[f, g](c):=\widehat{\mathcal{L}}\left(h^c \times f, g\right) .$	$[f, g](c):=\widehat{\mathcal{L}}\left(h^c \times f, g\right) .$	$[f, g](c):=\widehat{\mathcal{L}}\left(h^c \times f, g\right) .$
2106.07890_E00042	021	■ 1.000 ● N +0	$\widehat{\mathcal{L}}\left(h \times f, g\right)=\widehat{\mathcal{L}}\left(h,[f, g]\right) .$	$\widehat{\mathcal{L}}\left(h \times f, g\right)=\widehat{\mathcal{L}}\left(h,[f, g]\right) .$	$\widehat{\mathcal{L}}\left(h \times f, g\right)=\widehat{\mathcal{L}}\left(h,[f, g]\right) .$
2106.07890_E00043 (21)	021	■ 1.000 ● N +0	$\left[h^x, h^y\right](c)=\inf _{d \in \mathcal{L}}\left\{\frac{\pi(d \mid y)}{\min \left\{\pi(d \mid c), \pi(d \mid x)\right\}}, 1\right\} .$	$\left[h^x, h^y\right](c)=\inf _{d \in \mathcal{L}}\left\{\frac{\pi(d \mid y)}{\min \left\{\pi(d \mid c), \pi(d \mid x)\right\}}, 1\right\} .$	$\left[h^x, h^y\right](c)=\inf _{d \in \mathcal{L}}\left\{\frac{\pi(d \mid y)}{\min \left\{\pi(d \mid c), \pi(d \mid x)\right\}}, 1\right\} .$
2106.07890_E00045	023	■ 1.000 ● N -1	$\widehat{\mathcal{M}}(f, g)=\sup _{x \in \mathcal{M}} d_{[0, \infty]}(f x, g x)=\sup _{x \in \mathcal{M}} \max \{g x-f x, 0\}$	$\widehat{\mathcal{M}}(f, g)=\sup _{x \in \mathcal{M}} d_{[0, \infty]}(f x, g x)=\sup _{x \in \mathcal{M}} \max \{g x-f x, 0\}$	$\widehat{\mathcal{M}}(f, g)=\sup _{x \in \mathcal{M}} d_{[0, \infty]}(f x, g x)=\sup _{x \in \mathcal{M}} \max \{g x-f x, 0\}$
2106.07890_E00046	023	■ 1.000 ● W +1	$\begin{gathered} \left(\left(w_1, f\right) \times\left(w_2, g\right)\right)(c)=\max \left\{f c-w_1, g c-w_2, 0\right\} \\ \left(\left(w_1, f\right) \sqcup\left(w_2, g\right)\right)(c)=\min \left\{f c+w_1, g c+w_2\right\} \end{gathered}$	$\begin{gathered} \left(\left(w_1, f\right) \times\left(w_2, g\right)\right)(c)=\max \left\{f c-w_1, g c-w_2, 0\right\} \\ \left(\left(w_1, f\right) \sqcup\left(w_2, g\right)\right)(c)=\min \left\{f c+w_1, g c+w_2\right\} \end{gathered}$	$\begin{gathered} \left(\left(w_1, f\right) \times\left(w_2, g\right)\right)(c)=\max \left\{f c-w_1, g c-w_2, 0\right\} \\ \left(\left(w_1, f\right) \sqcup\left(w_2, g\right)\right)(c)=\min \left\{f c+w_1, g c+w_2\right\} \end{gathered}$
2106.07890_E00047	023	■ 1.000 ● N +0	$\left(\left(w_1^{\prime}, f^{\prime}\right) \times\left(w_2^{\prime}, g^{\prime}\right)\right)(c)=\min \left\{\frac{f^{\prime} c}{w_1^{\prime}}, \frac{g^{\prime} c}{w_2^{\prime}}, 1\right\}$	$\left(\left(w_1^{\prime}, f^{\prime}\right) \times\left(w_2^{\prime}, g^{\prime}\right)\right)(c)=\min \left\{\frac{f^{\prime} c}{w_1^{\prime}}, \frac{g^{\prime} c}{w_2^{\prime}}, 1\right\}$	$\left(\left(w_1^{\prime}, f^{\prime}\right) \times\left(w_2^{\prime}, g^{\prime}\right)\right)(c)=\min \left\{\frac{f^{\prime} c}{w_1^{\prime}}, \frac{g^{\prime} c}{w_2^{\prime}}, 1\right\}$
2106.07890_E00049	024	■ 1.000 ● C +5	$\begin{aligned} & \left(\left(w_1, f\right) \sqcup\left(w_2, g\right)\right)(c)=-\ln \left(\left(\left(w_1^{\prime}, f^{\prime}\right) \sqcup\left(w_2^{\prime}, g^{\prime}\right)\right)(c)\right) \\ & =-\ln \left(\max \left\{w_1^{\prime} f^{\prime} c, w_2^{\prime} g^{\prime} c\right\}\right) \\ & =\min \left\{-\ln \left(w_1^{\prime} f^{\prime} c\right),-\ln \left(w_2^{\prime} g^{\prime} c\right)\right\} \\ & =\min \left\{-\ln \left(f^{\prime} c\right)-\ln \left(w_1^{\prime}\right),-\ln \left(g^{\prime} c\right)-\ln \left(w_2^{\prime}\right)\right\} \\ & =\min \left\{f c+w_1, g c+w_2\right\} . \end{aligned}$	$\begin{aligned} & \left(\left(w_1, f\right) \sqcup\left(w_2, g\right)\right)(c)=-\ln \left(\left(\left(w_1^{\prime}, f^{\prime}\right) \sqcup\left(w_2^{\prime}, g^{\prime}\right)\right)(c)\right) \\ & =-\ln \left(\max \left\{w_1^{\prime} f^{\prime} c, w_2^{\prime} g^{\prime} c\right\}\right) \\ & =\min \left\{-\ln \left(w_1^{\prime} f^{\prime} c\right),-\ln \left(w_2^{\prime} g^{\prime} c\right)\right\} \\ & =\min \left\{-\ln \left(f^{\prime} c\right)-\ln \left(w_1^{\prime}\right),-\ln \left(g^{\prime} c\right)-\ln \left(w_2^{\prime}\right)\right\} \\ & =\min \left\{f c+w_1, g c+w_2\right\} . \end{aligned}$	$\begin{aligned} & \left(\left(w_1, f\right) \sqcup\left(w_2, g\right)\right)(c)=-\ln \left(\left(\left(w_1^{\prime}, f^{\prime}\right) \sqcup\left(w_2^{\prime}, g^{\prime}\right)\right)(c)\right) \\ & =-\ln \left(\max \left\{w_1^{\prime} f^{\prime} c, w_2^{\prime} g^{\prime} c\right\}\right) \\ & =\min \left\{-\ln \left(w_1^{\prime} f^{\prime} c\right),-\ln \left(w_2^{\prime} g^{\prime} c\right)\right\} \\ & =\min \left\{-\ln \left(f^{\prime} c\right)-\ln \left(w_1^{\prime}\right),-\ln \left(g^{\prime} c\right)-\ln \left(w_2^{\prime}\right)\right\} \\ & =\min \left\{f c+w_1, g c+w_2\right\} . \end{aligned}$
2106.07890_E00050	024	■ 1.000 ● N +0	$s_1 \oplus s_2=\min \left\{s_1, s_2\right\} \text { and } s_1 \odot s_2=s_1+s_2$	$s_1 \oplus s_2=\min \left\{s_1, s_2\right\} \text { and } s_1 \odot s_2=s_1+s_2$	$s_1 \oplus s_2=\min \left\{s_1, s_2\right\} \text { and } s_1 \odot s_2=s_1+s_2$
2106.07890_E00051	025	■ 1.000 ● N +0	$\left(\left(w_1, f\right) \sqcup\left(w_2, g\right)\right)(c)=\min \left\{f c+w_1, g c+w_2\right\}=w_1 \odot f c \oplus w_2 \odot g c .$	$\left(\left(w_1, f\right) \sqcup\left(w_2, g\right)\right)(c)=\min \left\{f c+w_1, g c+w_2\right\}=w_1 \odot f c \oplus w_2 \odot g c .$	$\left(\left(w_1, f\right) \sqcup\left(w_2, g\right)\right)(c)=\min \left\{f c+w_1, g c+w_2\right\}=w_1 \odot f c \oplus w_2 \odot g c .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered
2106.07890_F00001	003	—	$\mathcal{L}$	$\mathcal{L}$
2106.07890_F00002	004	—	$\widehat{\mathcal{L}}$	$\widehat{\mathcal{L}}$
2106.07890_F00003	002	—	$\{ \ }^{\{+ \}}$	$\{ \ }^{\{+ \}}$
2106.07890_F00004	002	—	$\mathrm{B}^{\{+ \}}$	$\mathrm{B}^{\{+ \}}$
2106.07890_F00005	003	—	∞	∞
2106.07890_F00006	005	—	$x \rightarrow y$	$x \rightarrow y$
2106.07890_F00007	005	—	x	x
2106.07890_F00008	005	—	y	y
2106.07890_F00009	005	—	z	z
2106.07890_F00010	005	—	$h^{\{x \}} := \operatorname{hom}(x, -)$	$h^x := \operatorname{hom}(x, -)$
2106.07890_F00011	005	—	$*$	$*$
2106.07890_F00012	005	—	$h^{\{x \}}$	h^x
2106.07890_F00013	005	—	$\widehat{\mathrm{C}} :=$	$\widehat{\mathrm{C}} :=$
2106.07890_F00014	005	—	$^{\{\mathrm{C} \}}$	C
2106.07890_F00015	006	—	$h^{\{x \}} \sqcup h^{\{y \}}$	$h^x \sqcup h^y$
2106.07890_F00016	006	—	$x =$	$x =$
2106.07890_F00017	006	—	$y =$	$y =$
2106.07890_F00018	006	—	$h^{\{\text{red} \}} \sqcup h^{\{\text{blue} \}}$	$h^{\text{red}} \sqcup h^{\text{blue}}$
2106.07890_F00019	006	—	c	c
2106.07890_F00020	006	—	$h^{\{\text{red} \}}(c) \sqcup h^{\{\text{blue} \}}(c)$	$h^{\text{red}}(c) \sqcup h^{\text{blue}}(c)$
2106.07890_F00021	006	—	$h^{\{x \}} \times h^{\{y \}} : \mathrm{C} \rightarrow$	$h^x \times h^y : \mathrm{C} \rightarrow$
2106.07890_F00022	006	—	$h^{\{\text{red} \}}(c) \times h^{\{\text{blue} \}}(c)$	$h^{\text{red}}(c) \times h^{\text{blue}}(c)$
2106.07890_F00023	006	—	$h^{\{\text{red} \}} \times h^{\{\text{blue} \}}$	$h^{\text{red}} \times h^{\text{blue}}$
2106.07890_F00024	006	—	$\widehat{\mathrm{C}}$	$\widehat{\mathrm{C}}$
2106.07890_F00025	006	—	$[\] : , \widehat{\mathrm{C}} \times \widehat{\mathrm{C}} \rightarrow \widehat{\mathrm{C}}$	$[\] : , \widehat{\mathrm{C}} \times \widehat{\mathrm{C}} \rightarrow \widehat{\mathrm{C}}$
2106.07890_F00026	006	—	$\widehat{\mathrm{C}}(F \times$	$\widehat{\mathrm{C}}(F \times$
2106.07890_F00027	006	—	$G, H) \cong \widehat{\mathrm{C}}(F, [G, H])$	$G, H) \cong \widehat{\mathrm{C}}(F, [G, H])$
2106.07890_F00028	006	—	F, G	F, G
2106.07890_F00029	006	—	H	H
2106.07890_F00030	006	—	$\mathrm{A}(x, y)$	$\mathrm{A}(x, y)$
2106.07890_F00031	006	—	$[G, H]$	$[G, H]$
2106.07890_F00032	006	—	$x \leq y$	$x \leq y$
2106.07890_F00033	006	—	$\wedge$	$\wedge$
2106.07890_F00034	006	—	$x \wedge y \leq z$	$x \wedge y \leq z$
2106.07890_F00035	006	—	$x \leq (y \Rightarrow z)$	$x \leq (y \Rightarrow z)$
2106.07890_F00036	006	—	x, y, z	x, y, z
2106.07890_F00037	007	—	$h^{\{c \}} \times h^{\{\text{red} \}}$	$h^c \times h^{\text{red}}$
2106.07890_F00038	007	—	$h^{\{\text{blue} \}}$	h^{blue}
2106.07890_F00039	007	—	$\left(h^{\{c \}} \times h^{\{\text{red} \}} \right) (d)$	$\left(h^c \times h^{\text{red}} \right) (d)$
2106.07890_F00040	007	—	$\left(h^{\{\text{blue} \}} \right) (d)$	$\left(h^{\text{blue}} \right) (d)$
2106.07890_F00041	007	—	$\left[h^{\{\text{red} \}}, h^{\{\text{blue} \}} \right] (c)$	$\left[h^{\text{red}}, h^{\text{blue}} \right] (c)$
2106.07890_F00042	007	—	$\emptyset$	$\emptyset$
2106.07890_F00043	007	—	$h^{\{c \}} \times$	$h^c \times$
2106.07890_F00044	007	—	$h^{\{\text{red} \}}(d) = h^{\{c \}}(d) \times h^{\{\text{red} \}}(d)$	$h^{\text{red}}(d) = h^c(d) \times h^{\text{red}}(d)$
2106.07890_F00045	007	—	d	d
2106.07890_F00046	007	—	$h^{\{\text{blue} \}}(d)$	$h^{\text{blue}}(d)$
2106.07890_F00047	007	—	$\left(h^{\{c \}} \times h^{\{\text{red} \}} \right) (d) = *$	$\left(h^c \times h^{\text{red}} \right) (d) = *$
2106.07890_F00048	007	—	$h^{\{\text{blue} \}}(d) = \emptyset$	$h^{\text{blue}}(d) = \emptyset$
2106.07890_F00049	007	—	$\left(h^{\{c \}} \times h^{\{\text{red} \}} \right) (d) \rightarrow h^{\{\text{blue} \}}(d)$	$\left(h^c \times h^{\text{red}} \right) (d) \rightarrow h^{\text{blue}}(d)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
2106.07890_F00050	007	—	$\left[h^{\text{red}}, h^{\text{blue}}\right]$	$\left[h^{\text{red}}, h^{\text{blue}}\right]$
2106.07890_F00051	007	—	$=*$	$= *$
2106.07890_F00052	007	—	$\left[h^{\text{red}}, h^{\text{blue}}\right]($	$\left[h^{\text{red}}, h^{\text{blue}}\right]($
2106.07890_F00053	007	—	$)=\emptyset$	$) = \emptyset$
2106.07890_F00054	007	—	$\left[h^x, h^y\right]: \mathrm{C} \rightarrow$	$\left[h^x, h^y\right]: \mathrm{C} \rightarrow$
2106.07890_F00055	008	—	$\mathrm{C}^{\mathrm{op}} \rightarrow$	$\mathrm{C}^{\mathrm{op}} \rightarrow$
2106.07890_F00056	008	—	$h^x=\operatorname{hom}(x,-)$	$h^x = \operatorname{hom}(x, -)$
2106.07890_F00057	008	—	C^{op}	C^{op}
2106.07890_F00058	008	—	$\mathrm{C}^{\mathrm{op}}(x,y):=\mathrm{C}(y,x)$	$\mathrm{C}^{\mathrm{op}}(x,y) := \mathrm{C}(y,x)$
2106.07890_F00059	008	—	$\mathrm{C}(x,y)$	$\mathrm{C}(x,y)$
2106.07890_F00060	008	—	$[0,1]$	$[0,1]$
2106.07890_F00061	008	—	a	a
2106.07890_F00062	008	—	b	b
2106.07890_F00063	009	—	$(\mathcal{V}, \leq, \otimes, 1)$	$(\mathcal{V}, \leq, \otimes, 1)$
2106.07890_F00064	009	—	$(\mathcal{V}, \leq)$	$(\mathcal{V}, \leq)$
2106.07890_F00065	009	—	$(\mathcal{V}, \otimes, 1)$	$(\mathcal{V}, \otimes, 1)$
2106.07890_F00066	009	—	$x \otimes y \leq x^{\prime} \otimes y^{\prime}$	$x \otimes y \leq x' \otimes y'$
2106.07890_F00067	009	—	$x \leq x^{\prime}$	$x \leq x'$
2106.07890_F00068	009	—	$y \leq y^{\prime}$	$y \leq y'$
2106.07890_F00069	009	—	$\mathcal{V}, \leq, \otimes, 1$	$\mathcal{V}, \leq, \otimes, 1$
2106.07890_F00070	009	—	$\mathcal{V}$	$\mathcal{V}$
2106.07890_F00071	009	—	$\mathcal{C}$	$\mathcal{C}$
2106.07890_F00072	009	—	$\mathcal{C}(x,y) \in \mathcal{V}$	$\mathcal{C}(x,y) \in \mathcal{V}$
2106.07890_F00073	009	—	$x,y,z \in \mathcal{C}$	$x,y,z \in \mathcal{C}$
2106.07890_F00074	009	—	$[0,1]:=\{x \in \mathbb{R}: \, 0 \leq x \leq 1\}$	$[0,1] := \{x \in \mathbb{R} : 0 \leq x \leq 1\}$
2106.07890_F00075	009	—	$\leq$	$\leq$
2106.07890_F00076	009	—	$(x,y) \mapsto \mathcal{C}(x,y)$	$(x,y) \mapsto \mathcal{C}(x,y)$
2106.07890_F00077	009	—	$x,y \in \mathcal{C}$	$x,y \in \mathcal{C}$
2106.07890_F00078	009	—	$\mathcal{C}(x,x)=1$	$\mathcal{C}(x,x) = 1$
2106.07890_F00079	009	—	$x \in \mathcal{C}$	$x \in \mathcal{C}$
2106.07890_F00080	009	—	$\mathcal{C}(y,z)\mathcal{C}(x,y) \leq \mathcal{C}(x,z)$	$\mathcal{C}(y,z)\mathcal{C}(x,y) \leq \mathcal{C}(x,z)$
2106.07890_F00081	009	—	$[x,y] \in \mathcal{V}$	$[x,y] \in \mathcal{V}$
2106.07890_F00082	009	—	$\mathcal{V}(x,y)$	$\mathcal{V}(x,y)$
2106.07890_F00083	010	—	$[x,y]$	$[x,y]$
2106.07890_F00084	010	—	$(x,y) \mapsto [x,y]$	$(x,y) \mapsto [x,y]$
2106.07890_F00085	010	—	$1 \leq [x,x]$	$1 \leq [x,x]$
2106.07890_F00086	010	—	$[y,z] \otimes [x,y] \leq [x,z]$	$[y,z] \otimes [x,y] \leq [x,z]$
2106.07890_F00087	010	—	$1 \otimes x \leq x$	$1 \otimes x \leq x$
2106.07890_F00088	010	—	$[y,z] \leq [y,z]$	$[y,z] \leq [y,z]$
2106.07890_F00089	010	—	$[y,z] \otimes y \leq z$	$[y,z] \otimes y \leq z$
2106.07890_F00090	010	—	$[x,y] \otimes x \leq y$	$[x,y] \otimes x \leq y$
2106.07890_F00091	010	—	$[y,z] \otimes [x,y] \otimes x \leq z$	$[y,z] \otimes [x,y] \otimes x \leq z$
2106.07890_F00092	010	—	$a \otimes b:=a \, b$	$a \otimes b := ab$
2106.07890_F00093	010	—	$a,b \in [0,1]$	$a,b \in [0,1]$
2106.07890_F00094	010	—	$ab \leq c$	$ab \leq c$
2106.07890_F00095	010	—	$a \leq [b,c]$	$a \leq [b,c]$
2106.07890_F00096	010	—	$c<b$	$c < b$
2106.07890_F00097	010	—	$a \leq \frac{c}{b} = [b,c]$	$a \leq \frac{c}{b} = [b,c]$
2106.07890_F00098	010	—	$b \leq c$	$b \leq c$
2106.07890_F00099	010	—	$[b,c]=1$	$[b,c] = 1$
2106.07890_F00100		—	$\square$	$\square$
2106.07890_F00101	010	—	$a,b,c \in [0,1]$	$a,b,c \in [0,1]$
2106.07890_F00102	010	—	$\min \{b \, / \, a, 1\}$	$\min\{b/a, 1\}$
2106.07890_F00103	010	—	$b \, / \, \theta \geq 1$	$b/0 \geq 1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
2106.07890_F00104	010	—	$\emptyset \leq b \leq 1$	$0 \leq b \leq 1$
2106.07890_F00105	010	—	$a\,b$	ab
2106.07890_F00106	010	—	$a \,\otimes\, b$	$a \otimes b$
2106.07890_F00107	010	—	$a \,\times\, b$	$a \times b$
2106.07890_F00108	010	—	$\min\{a,b\}$	$\min\{a,b\}$
2106.07890_F00109	010	—	$a \,\sqcup\, b$	$a \sqcup b$
2106.07890_F00110	010	—	$\mathrm{C}(x,y)=1$	$\mathcal{C}(x,y)=1$
2106.07890_F00111	010	—	$\mathrm{C}(x,y)=0$	$\mathcal{C}(x,y)=0$
2106.07890_F00112	010	—	$2:=\{0,1\}$	$2 := \{0,1\}$
2106.07890_F00113	011	—	$\pi(y \,\mid\, x)$	$\pi(y \mid x)$
2106.07890_F00114	011	—	$\pi(y \,\mid\, x)=0$	$\pi(y \mid x) = 0$
2106.07890_F00115	011	—	$\pi(x \,\mid\, x)=$	$\pi(x \mid x) =$
2106.07890_F00116	011	—	$\mathcal{D}$	$\mathcal{D}$
2106.07890_F00117	012	—	$\mathcal{L} \rightarrow \widehat{\mathcal{L}}$	$\mathcal{L} \rightarrow \widehat{\mathcal{L}}$
2106.07890_F00118	012	—	$\mathcal{L} \rightarrow \mathcal{D}$	$\mathcal{L} \rightarrow \mathcal{D}$
2106.07890_F00119	012	—	$\mathcal{D} \cdots \cdots > \widehat{\mathcal{L}}$	$\mathcal{D} \cdots \cdots > \widehat{\mathcal{L}}$
2106.07890_F00120	012	—	$(\mathcal{V}, \otimes, \leq, 1)$	$(\mathcal{V}, \otimes, \leq, 1)$
2106.07890_F00121	012	—	$\mathcal{C} \rightarrow \mathcal{D}$	$\mathcal{C} \rightarrow \mathcal{D}$
2106.07890_F00122	012	—	$f: \mathcal{C} \rightarrow \mathcal{D}$	$f: \mathcal{C} \rightarrow \mathcal{D}$
2106.07890_F00123	012	—	$\mathcal{D}=\mathcal{V}$	$\mathcal{D} = \mathcal{V}$
2106.07890_F00124	012	—	$f: \mathcal{C} \rightarrow \mathcal{V}$	$f: \mathcal{C} \rightarrow \mathcal{V}$
2106.07890_F00125	012	—	$\mathcal{C}(x,y) \leq$	$\mathcal{C}(x,y) \leq$
2106.07890_F00126	012	—	$\mathcal{V}(fx,fy)$	$\mathcal{V}(fx,fy)$
2106.07890_F00127	012	—	$\mathcal{D}^{\mathcal{C}}$	$\mathcal{D}^{\mathcal{C}}$
2106.07890_F00128	012	—	$\mathcal{D}^{\mathcal{C}}(f,g) \in \mathcal{V}$	$\mathcal{D}^{\mathcal{C}}(f,g) \in \mathcal{V}$
2106.07890_F00129	012	—	$f,g: \mathcal{C} \rightarrow \mathcal{D}$	$f,g: \mathcal{C} \rightarrow \mathcal{D}$
2106.07890_F00130	013	—	$\widehat{\mathcal{C}} := [0,1]^{\mathcal{C}}$	$\widehat{\mathcal{C}} := [0,1]^{\mathcal{C}}$
2106.07890_F00131	013	—	$f,g: \mathcal{C} \rightarrow [0,1]$	$f,g: \mathcal{C} \rightarrow [0,1]$
2106.07890_F00132	013	—	$\widehat{\mathcal{C}}(f,g)=\inf_{c \in \mathcal{C}}\{1,g c / f c\}$	$\widehat{\mathcal{C}}(f,g)=\inf_{c \in \mathcal{C}}\{1,g c / f c\}$
2106.07890_F00133	013	—	$\mathcal{V}=[0,1]$	$\mathcal{V} = [0,1]$
2106.07890_F00134	013	—	$\mathcal{D}=[0,1]$	$\mathcal{D} = [0,1]$
2106.07890_F00135	013	—	$\mathcal{C} \rightarrow [0,1]$	$\mathcal{C} \rightarrow [0,1]$
2106.07890_F00136	013	—	$c,d \in \mathcal{C}$	$c,d \in \mathcal{C}$
2106.07890_F00137	013	—	$\mathcal{C}(c,d) \leq [\mathcal{C}(x,c),\mathcal{C}(x,d)]$	$\mathcal{C}(c,d) \leq [\mathcal{C}(x,c),\mathcal{C}(x,d)]$
2106.07890_F00138	013	—	$\mathcal{C}(c,d)\mathcal{C}(x,c) \leq \mathcal{C}(x,d)$	$\mathcal{C}(c,d)\mathcal{C}(x,c) \leq \mathcal{C}(x,d)$
2106.07890_F00139	013	—	$h^x :=$	$h^x :=$
2106.07890_F00140	013	—	$\mathcal{C}(x,-)$	$\mathcal{C}(x,-)$
2106.07890_F00141	013	—	$f: \mathcal{C} \rightarrow [0,1]$	$f: \mathcal{C} \rightarrow [0,1]$
2106.07890_F00142	013	—	$f=h^x$	$f = h^x$
2106.07890_F00143	013	—	$x \mapsto h^x$	$x \mapsto h^x$
2106.07890_F00144	013	—	$\widehat{\mathcal{C}}(h^x,f)=f(x)$	$\widehat{\mathcal{C}}(h^x,f)=f(x)$
2106.07890_F00145	013	—	f	f
2106.07890_F00146	013	—	$\widehat{\mathcal{C}}(h^x,f)=\inf_{c \in \mathcal{C}}\{\left[h^x(c),fc\right]\}$	$\widehat{\mathcal{C}}(h^x,f)=\inf_{c \in \mathcal{C}}\{[h^x(c),fc]\}$
2106.07890_F00147	013	—	$c \in \mathcal{C}$	$c \in \mathcal{C}$
2106.07890_F00148	013	—	$\widehat{\mathcal{C}}(h^x,f) \leq [h^x(c),fc]$	$\widehat{\mathcal{C}}(h^x,f) \leq [h^x(c),fc]$
2106.07890_F00149	013	—	$c=x$	$c = x$
2106.07890_F00150	014	—	$\mathcal{C}(x,c) \leq$	$\mathcal{C}(x,c) \leq$
2106.07890_F00151	014	—	$[fx,fc]$	$[fx,fc]$
2106.07890_F00152	014	—	$\mathcal{C}(x,c) \leq [fx,fc]$	$\mathcal{C}(x,c) \leq [fx,fc]$
2106.07890_F00153	014	—	$\mathcal{C}(x,c) f x \leq f c$	$\mathcal{C}(x,c) f x \leq f c$
2106.07890_F00154	014	—	$f x \leq [\mathcal{C}(x,c),fc]$	$f x \leq [\mathcal{C}(x,c),fc]$
2106.07890_F00155	014	—	$f x \leq \inf_{c \in \mathcal{C}}\{\left[h^x(c),fc\right]\}$	$f x \leq \inf_{c \in \mathcal{C}}\{[h^x(c),fc]\} =$
2106.07890_F00156	014	—	$\widehat{\mathcal{C}}(h^x,f)$	$\widehat{\mathcal{C}}(h^x,f)$
2106.07890_F00157	014	—	$\mathcal{C}(y,x)=\widehat{\mathcal{C}}(h^x,h^y)$	$\mathcal{C}(y,x)=\widehat{\mathcal{C}}(h^x,h^y)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
2106.07890_F00158	014	—	x, y	x, y
2106.07890_F00159	014	—	$f=h^{\{y\}}$	$f = h^y$
2106.07890_F00160	014	—	$\widehat{\mathrm{cal}\{C\}}\left(h^{\{x\}}, h^{\{y\}}\right)=h^{\{y\}}(x)=\mathrm{cal}\{C\}(y, x)$	$\hat{\mathcal{C}}(h^x, h^y) = h^y(x) = \mathcal{C}(y, x)$
2106.07890_F00161	014	—	$\mathrm{cal}\{C\}^{\{o\ p\}} \rightarrow \widehat{\mathrm{cal}\{C\}}$	$\mathcal{C}^{op} \rightarrow \hat{\mathcal{C}}$
2106.07890_F00162	014	—	$\mathrm{cal}\{C\}^{\{o\ p\}}$	$\mathcal{C}^{op}$
2106.07890_F00163	014	—	$\widehat{\mathrm{cal}\{C\}}$	$\hat{\mathcal{C}}$
2106.07890_F00164	014	—	$o\ p$	op
2106.07890_F00165	014	—	$\mathrm{cal}\{C\}^{\{\mathrm{op}\}}$	$\mathcal{C}^{\mathrm{op}}$
2106.07890_F00166	014	—	Y	Y
2106.07890_F00167	014	—	$\{X \rightarrow Y\}$	$\{X \rightarrow Y\}$
2106.07890_F00168	014	—	X	X
2106.07890_F00169	014	—	$\widehat{\mathrm{cal}\{C\}}=[0,1]^{\{\mathrm{cal}\{C\}}}$	$\hat{\mathcal{C}} = [0, 1]^{\mathcal{C}}$
2106.07890_F00170	014	—	$\widehat{\mathrm{cal}\{L\}}:=[0,1]^{\{\mathrm{cal}\{L\}}}$	$\hat{\mathcal{L}} := [0, 1]^{\mathcal{L}}$
2106.07890_F00171	014	—	$h^{\{x\}}:=\mathrm{cal}\{L\}(x,-)$	$h^x := \mathcal{L}(x, -)$
2106.07890_F00172	014	—	$x \leq c$	$x \leq c$
2106.07890_F00173	015	—	$\mathrm{cal}\{L\}(x, y)=a \neq 0$	$\mathcal{L}(x, y) = a \neq 0$
2106.07890_F00174	015	—	$h^{\{y\}}$	h^y
2106.07890_F00175	015	—	$\{ \ }^{\{\mathrm{cal}\{C\}}}$	$\mathcal{C}$
2106.07890_F00176	016	—	$F: \mathrm{cal}\{J\} \rightarrow \mathrm{cal}\{C\}$	$F: \mathcal{J} \rightarrow \mathcal{C}$
2106.07890_F00177	016	—	$\lim F$	$\lim F$
2106.07890_F00178	016	—	Z	Z
2106.07890_F00179	016	—	$\mathrm{cal}\{C\}(Z, \lim F) \cong \operatorname{Set}^{\{\mathrm{cal}\{J\}}(*, \mathrm{cal}\{C\}(Z, F))$	$\mathcal{C}(Z, \lim F) \cong \operatorname{Set}^{\mathcal{J}}(*, \mathcal{C}(Z, F))$
2106.07890_F00180	016	—	(Z, F)	(Z, F)
2106.07890_F00181	016	—	$*=*(i) \rightarrow \mathrm{cal}\{C\}(Z, F\ i)$	$* = *(i) \rightarrow \mathcal{C}(Z, Fi)$
2106.07890_F00182	016	—	i	i
2106.07890_F00183	016	—	$Z \rightarrow F\ i$	$Z \rightarrow Fi$
2106.07890_F00184	016	—	$W: \mathrm{cal}\{J\} \rightarrow \mathrm{cal}\{V\}$	$W: \mathcal{J} \rightarrow \mathcal{V}$
2106.07890_F00185	016	—	$\mathrm{cal}\{J\}$	$\mathcal{J}$
2106.07890_F00186	016	—	$\mathrm{cal}\{E\}$	$\mathcal{E}$
2106.07890_F00187	016	—	$F: \mathrm{cal}\{J\} \rightarrow \mathrm{cal}\{E\}$	$F: \mathcal{J} \rightarrow \mathcal{E}$
2106.07890_F00188	016	—	F	F
2106.07890_F00189	016	—	$\lim_{\{W\}} F$	$\lim_{\mathcal{W}} F$
2106.07890_F00190	016	—	$\mathrm{cal}\{E\} \rightarrow \mathrm{cal}\{V\}$	$\mathcal{E} \rightarrow \mathcal{V}$
2106.07890_F00191	017	—	$\mathrm{cal}\{E\}\left(Z, \lim_{\{W\}} F\right) \cong$	$\mathcal{E}\left(Z, \lim_{\mathcal{W}} F\right) \cong$
2106.07890_F00192	017	—	$\mathrm{cal}\{V\}^{\{\mathrm{cal}\{J\}}(W, \mathrm{cal}\{E\}(Z, F))$	$\mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(Z, F))$
2106.07890_F00193	017	—	$W: \mathrm{cal}\{J\}^{\{\mathrm{cal}\{op\}} \rightarrow \mathrm{cal}\{V\}$	$W: \mathcal{J}^{\mathrm{op}} \rightarrow \mathcal{V}$
2106.07890_F00194	017	—	$\operatorname{colim}_{\{W\}} F$	$\operatorname{colim}_{\mathcal{W}} F$
2106.07890_F00195	017	—	$\mathrm{cal}\{E\}=\widehat{\mathrm{cal}\{L\}}:=[0,1]^{\{\mathrm{cal}\{L\}}}$	$\mathcal{E} = \hat{\mathcal{L}} := [0, 1]^{\mathcal{L}}$
2106.07890_F00196	017	—	$\mathrm{cal}\{J\}(i, j)=\delta_{i\ j}$	$\mathcal{J}(i, j) = \delta_{ij}$
2106.07890_F00197	017	—	$i, j \in \{1, 2\}$	$i, j \in \{1, 2\}$
2106.07890_F00198	017	—	$W: \mathrm{cal}\{J\} \rightarrow$	$W: \mathcal{J} \rightarrow$
2106.07890_F00199	017	—	$w_{\{1\}}:=W(1)$	$w_1 := W(1)$
2106.07890_F00200	017	—	$w_{\{2\}}:=$	$w_2 :=$
2106.07890_F00201	017	—	$W(2)$	$W(2)$
2106.07890_F00202	017	—	$f, g: \mathrm{cal}\{L\} \rightarrow [0, 1]$	$f, g: \mathcal{L} \rightarrow [0, 1]$
2106.07890_F00203	017	—	$F: \mathrm{cal}\{J\} \rightarrow$	$F: \mathcal{J} \rightarrow$
2106.07890_F00204	017	—	$f:=F(1)$	$f := F(1)$
2106.07890_F00205	017	—	$g:=F(2)$	$g := F(2)$
2106.07890_F00206	017	—	W	W
2106.07890_F00207	017	—	$\left(w_{\{1\}}, f\right) \times \left(w_{\{2\}}, g\right): \mathrm{cal}\{L\} \rightarrow [0, 1]$	$(w_1, f) \times (w_2, g): \mathcal{L} \rightarrow [0, 1]$
2106.07890_F00208	017	—	$c \mapsto$	$c \mapsto$
2106.07890_F00209	017	—	$\min \left\{ \frac{f\ c}{w_1}, \frac{g\ c}{w_2}, 1 \right\}$	$\min \left\{ \frac{fc}{w_1}, \frac{gc}{w_2}, 1 \right\}$
2106.07890_F00210	017	—	$c \mapsto \min \left\{ \frac{f\ c}{w_1}, \frac{g\ c}{w_2}, 1 \right\}$	$c \mapsto \min \left\{ \frac{fc}{w_1}, \frac{gc}{w_2}, 1 \right\}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
2106.07890_F00211	017	—	$Z: \mathcal{L} \rightarrow [0,1]$	$Z: \mathcal{L} \rightarrow [0,1]$
2106.07890_F00212	017	—	$\widehat{\mathcal{L}}(Z, F)$	$\widehat{\mathcal{L}}(Z, F)$
2106.07890_F00213	017	—	$J \rightarrow [0,1]$	$J \rightarrow [0,1]$
2106.07890_F00214	017	—	J	J
2106.07890_F00215	017	—	$[0,1]^{\mathcal{J}}$	$[0,1]^{\mathcal{J}}$
2106.07890_F00216	017	—	$f=F(1)$	$f=F(1)$
2106.07890_F00217	017	—	$g=F(2)$	$g=F(2)$
2106.07890_F00218	018	—	$\mathcal{L}(c, d) \leq \left[\lim_{\mathcal{W}} F(c), \lim_{\mathcal{W}} F(d) \right]$	$\mathcal{L}(c, d) \leq \left[\lim_{\mathcal{W}} F(c), \lim_{\mathcal{W}} F(d) \right]$
2106.07890_F00219	018	—	$\mathcal{L}(c, d) \lim_{\mathcal{W}} F(c) \leq$	$\mathcal{L}(c, d) \lim_{\mathcal{W}} F(c) \leq$
2106.07890_F00220	018	—	$\lim_{\mathcal{W}} F(d)$	$\lim_{\mathcal{W}} F(d)$
2106.07890_F00221	018	—	$c, d \in \mathcal{L}$	$c, d \in \mathcal{L}$
2106.07890_F00222	018	—	g	g
2106.07890_F00223	018	—	$(w_1, f) \times (w_2, g)$	$(w_1, f) \times (w_2, g)$
2106.07890_F00224	018	—	$w_1, w_2 \in [0,1]$	$w_1, w_2 \in [0,1]$
2106.07890_F00225	018	—	$(w_1, h^x) \times (w_2, h^y): \mathcal{L} \rightarrow [0,1]$	$(w_1, h^x) \times (w_2, h^y): \mathcal{L} \rightarrow [0,1]$
2106.07890_F00226	018	—	$\pi(c \mid x) \leq w_1$	$\pi(c \mid x) \leq w_1$
2106.07890_F00227	018	—	$\pi(c \mid y) \leq w_2$	$\pi(c \mid y) \leq w_2$
2106.07890_F00228	018	—	$w_1 h^x(c) \times w_2 h^y(c)$	$w_1 h^x(c) \times w_2 h^y(c)$
2106.07890_F00229	018	—	$(w_1, h^x) \times (w_2, h^y)$	$(w_1, h^x) \times (w_2, h^y)$
2106.07890_F00230	018	—	$w_1 = w_2 = 1$	$w_1 = w_2 = 1$
2106.07890_F00231	018	—	$f \times g$	$f \times g$
2106.07890_F00232	018	—	$f, g, h: \mathcal{L} \rightarrow [0,1]$	$f, g, h: \mathcal{L} \rightarrow [0,1]$
2106.07890_F00233	019	—	$W: \mathcal{J}^{\mathrm{op}} \rightarrow [0,1]$	$W: \mathcal{J}^{\mathrm{op}} \rightarrow [0,1]$
2106.07890_F00234	019	—	$w_2 := W(2)$	$w_2 := W(2)$
2106.07890_F00235	019	—	$F: \mathcal{J} \rightarrow \widehat{\mathcal{L}}$	$F: \mathcal{J} \rightarrow \widehat{\mathcal{L}}$
2106.07890_F00236	019	—	$(w_1, f) \sqcup (w_2, g): \mathcal{L} \rightarrow [0,1]$	$(w_1, f) \sqcup (w_2, g): \mathcal{L} \rightarrow [0,1]$
2106.07890_F00237	019	—	$\max\{w_1 f c, w_2 g c\}$	$\max\{w_1 f c, w_2 g c\}$
2106.07890_F00238	019	—	$\mathcal{J} = \mathcal{J}^{\mathrm{op}}$	$\mathcal{J} = \mathcal{J}^{\mathrm{op}}$
2106.07890_F00239	019	—	$\widehat{\mathcal{L}}(F-, Z)$	$\widehat{\mathcal{L}}(F-, Z)$
2106.07890_F00240	019	—	$i = 1, 2$	$i = 1, 2$
2106.07890_F00241	020	—	$c \mapsto \max\{w_1 f c, w_2 g c\}$	$c \mapsto \max\{w_1 f c, w_2 g c\}$
2106.07890_F00242	020	—	$\mathcal{L}(c, d) \leq [\mathrm{colim}_{\mathcal{W}} F(c), \mathrm{colim}_{\mathcal{W}} F(d)]$	$\mathcal{L}(c, d) \leq [\mathrm{colim}_{\mathcal{W}} F(c), \mathrm{colim}_{\mathcal{W}} F(d)]$
2106.07890_F00243	020	—	$\mathcal{L}(c, d) \mathrm{colim}_{\mathcal{W}} F(c) \leq \mathrm{colim}_{\mathcal{W}} F(d)$	$\mathcal{L}(c, d) \mathrm{colim}_{\mathcal{W}} F(c) \leq \mathrm{colim}_{\mathcal{W}} F(d)$
2106.07890_F00244	020	—	$(w_1, f) \sqcup (w_2, g)$	$(w_1, f) \sqcup (w_2, g)$
2106.07890_F00245	020	—	$G, H: \mathbf{C} \rightarrow$	$G, H: \mathbf{C} \rightarrow$
2106.07890_F00246	020	—	$c \mapsto \widehat{\mathbf{C}}(h^c \times G, H)$	$c \mapsto \widehat{\mathbf{C}}(h^c \times G, H)$
2106.07890_F00247	020	—	$h^c \times G$	$h^c \times G$
2106.07890_F00248	020	—	$[f, g]: \mathcal{L} \rightarrow [0,1]$	$[f, g]: \mathcal{L} \rightarrow [0,1]$
2106.07890_F00249	020	—	$h^c \times f$	$h^c \times f$
2106.07890_F00250	020	—	$\mathcal{L}(c, d) \leq [[f, g](c), [f, g](d)]$	$\mathcal{L}(c, d) \leq [[f, g](c), [f, g](d)]$
2106.07890_F00251	020	—	$\mathcal{L}(c, d)[f, g](c) \leq [f, g](d)$	$\mathcal{L}(c, d)[f, g](c) \leq [f, g](d)$
2106.07890_F00252	020	—	$\mathcal{L}(c, d) \leq [g(c), g(d)]$	$\mathcal{L}(c, d) \leq [g(c), g(d)]$
2106.07890_F00253	020	—	$\mathcal{L}(c, d)g(c) \leq g(d)$	$\mathcal{L}(c, d)g(c) \leq g(d)$
2106.07890_F00254	020	—	$\mathcal{L}(c, d) \widehat{\mathcal{L}}(h^c, g) \leq \widehat{\mathcal{L}}(h^d, g)$	$\mathcal{L}(c, d) \widehat{\mathcal{L}}(h^c, g) \leq \widehat{\mathcal{L}}(h^d, g)$
2106.07890_F00255	020	—	$\mathcal{L}(c, d) \left(\widehat{\mathcal{L}}(h^c, g) \times \widehat{\mathcal{L}}(f, g) \right) \leq \widehat{\mathcal{L}}(h^d, g) \times \widehat{\mathcal{L}}(f, g)$	$\mathcal{L}(c, d) \left(\widehat{\mathcal{L}}(h^c, g) \times \widehat{\mathcal{L}}(f, g) \right) \leq \widehat{\mathcal{L}}(h^d, g) \times \widehat{\mathcal{L}}(f, g)$
2106.07890_F00256	020	—	$\mathcal{L}(c, d) \widehat{\mathcal{L}}(h^c \times f, g) \leq \widehat{\mathcal{L}}(h^d \times f, g)$	$\mathcal{L}(c, d) \widehat{\mathcal{L}}(h^c \times f, g) \leq \widehat{\mathcal{L}}(h^d \times f, g)$
2106.07890_F00257	021	—	$\mathcal{L}(c, d) \leq \left[\widehat{\mathcal{L}}(h^c \times f, g), \widehat{\mathcal{L}}(h^d \times f, g) \right] = [[f, g](c), [f, g](d)]$	$\mathcal{L}(c, d) \leq \left[\widehat{\mathcal{L}}(h^c \times f, g), \widehat{\mathcal{L}}(h^d \times f, g) \right] = [[f, g](c), [f, g](d)]$
2106.07890_F00258	021	—	$(f \times -): \widehat{\mathcal{L}} \rightarrow \widehat{\mathcal{L}}$	$(f \times -): \widehat{\mathcal{L}} \rightarrow \widehat{\mathcal{L}}$
2106.07890_F00259	021	—	$[f, -]$	$[f, -]$
2106.07890_F00260	021	—	$- \times f$	$- \times f$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
2106.07890_F00261	021	—	$\widehat{\mathrm{cal}}\{L\}\left(h^c,[f,g]\right)=[f,g](c)$	$\widehat{\mathcal{L}}(h^c,[f,g])=[f,g](c)$
2106.07890_F00262	021	—	$[f,g](c)=\widehat{\mathrm{cal}}\{L\}\left(h^c\times f,g\right)$	$[f,g](c)=\widehat{\mathcal{L}}(h^c\times f,g)$
2106.07890_F00263	021	—	$\widehat{\mathrm{cal}}\{L\}\left(h^c,[f,g]\right)=\widehat{\mathrm{cal}}\{L\}\left(h^c\times f,g\right)$	$\widehat{\mathcal{L}}(h^c,[f,g])=\widehat{\mathcal{L}}(h^c\times f,g)$
2106.07890_F00264	021	—	h^c	h^c
2106.07890_F00265	021	—	h	h
2106.07890_F00266	021	—	$[f,g]$	$[f,g]$
2106.07890_F00267	021	—	$x,y\in\mathrm{cal}\{L\}$	$x,y\in\mathcal{L}$
2106.07890_F00268	021	—	$\left[h^x,h^y\right](c)=$	$[h^x,h^y](c)=$
2106.07890_F00269	021	—	$d=c$	$d=c$
2106.07890_F00270	021	—	$\pi(c\mid y)=$	$\pi(c\mid y)=$
2106.07890_F00271	021	—	$\pi(c\mid c)\times\pi(c\mid x)=\pi(c\mid x)\neq 0$	$\pi(c\mid c)\times\pi(c\mid x)=\pi(c\mid x)\neq 0$
2106.07890_F00272	021	—	$\left[h^x,h^y\right](c)\neq 0$	$[h^x,h^y](c)\neq 0$
2106.07890_F00273	021	—	$x\rightarrowtail y$	$x\Rightarrow y$
2106.07890_F00274	021	—	$\left[h^x,h^y\right]:\mathrm{cal}\{L\}\rightarrowtail[0,1]$	$[h^x,h^y]:\mathcal{L}\rightarrowtail[0,1]$
2106.07890_F00275	021	—	$\mathrm{cal}\{M\}$	$\mathcal{M}$
2106.07890_F00276	021	—	$[0,\infty]$	$[0,\infty]$
2106.07890_F00277	021	—	$a+\infty:=\infty$	$a+\infty:=\infty$
2106.07890_F00278	021	—	$\infty+a:=\infty$	$\infty+a:=\infty$
2106.07890_F00279	021	—	$b\leq a$	$b\leq a$
2106.07890_F00280	022	—	$d_{\mathrm{cal}\{M\}}$	$d_{\mathcal{M}}$
2106.07890_F00281	022	—	$[a,b]:=\max\{b-a,0\}$	$[a,b]:=\max\{b-a,0\}$
2106.07890_F00282	022	—	$a\times b=\max\{a,b\}$	$a\times b=\max\{a,b\}$
2106.07890_F00283	022	—	$a\sqcup b=\min\{a,b\}$	$a\sqcup b=\min\{a,b\}$
2106.07890_F00284	022	—	$-\ln:[0,1]\rightarrowtail[0,\infty]$	$-\ln:[0,1]\rightarrowtail[0,\infty]$
2106.07890_F00285	022	—	$[0,\infty]\rightarrowtail[0,1]$	$[0,\infty]\rightarrowtail[0,1]$
2106.07890_F00286	022	—	$a\mapsto\exp(-a)$	$a\mapsto\exp(-a)$
2106.07890_F00287	022	—	$\mathrm{cal}\{M\}(x,y):=-\ln\mathrm{cal}\{L\}(x,y)$	$\mathcal{M}(x,y):=-\ln\mathcal{L}(x,y)$
2106.07890_F00288	022	—	$0\geq\mathrm{cal}\{M\}(x,x)$	$0\geq\mathcal{M}(x,x)$
2106.07890_F00289	022	—	$\mathrm{cal}\{M\}(x,y)+\mathrm{cal}\{M\}(y,z)\geq\mathrm{cal}\{M\}(x,z)$	$\mathcal{M}(x,y)+\mathcal{M}(y,z)\geq\mathcal{M}(x,z)$
2106.07890_F00290	022	—	$d_{\mathrm{cal}\{M\}}(x,y):=\mathrm{cal}\{M\}(x,y)$	$d_{\mathcal{M}}(x,y):=\mathcal{M}(x,y)$
2106.07890_F00291	022	—	$\widehat{\mathrm{cal}}\{M\}:=[0,\infty]^{\mathrm{cal}\{M\}}$	$\widehat{\mathcal{M}}:=[0,\infty]^{\mathcal{M}}$
2106.07890_F00292	023	—	$f:\mathrm{cal}\{M\}\rightarrowtail[0,\infty]$	$f:\mathcal{M}\rightarrowtail[0,\infty]$
2106.07890_F00293	023	—	$d_{\mathrm{M}}(x,y)\geq[f x,f y]=\max\{f y-f x,0\}$	$d_M(x,y)\geq[f x,f y]=\max\{f y-f x,0\}$
2106.07890_F00294	023	—	$[a,b]=\max\{b-a,0\}$	$[a,b]=\max\{b-a,0\}$
2106.07890_F00295	023	—	$d_{\{0,\infty\}}(a,b)$	$d_{[0,\infty]}(a,b)$
2106.07890_F00296	023	—	$d_{\{0,\infty\}}(f x,f y)\leq d_{\mathrm{M}}(x,y)$	$d_{[0,\infty]}(f x,f y)\leq d_M(x,y)$
2106.07890_F00297	023	—	$\widehat{\mathrm{cal}}\{M\}$	$\widehat{\mathcal{M}}$
2106.07890_F00298	023	—	$\mathrm{cal}\{M\}^{\mathrm{op}}\rightarrowtail\widehat{\mathrm{cal}}\{M\}$	$\mathcal{M}^{\mathrm{op}}\rightarrowtail\widehat{\mathcal{M}}$
2106.07890_F00299	023	—	$x\mapsto d_{\mathrm{cal}\{M\}}(x,-)$	$x\mapsto d_{\mathcal{M}}(x,-)$
2106.07890_F00300	023	—	$f,g\in\widehat{\mathrm{cal}}\{M\}$	$f,g\in\widehat{\mathcal{M}}$
2106.07890_F00301	023	—	w_1,w_2	w_1,w_2
2106.07890_F00302	023	—	$f,g:\mathrm{cal}\{M\}\rightarrowtail[0,\infty]$	$f,g:\mathcal{M}\rightarrowtail[0,\infty]$
2106.07890_F00303	023	—	$w_1,w_2\in[0,\infty]$	$w_1,w_2\in[0,\infty]$
2106.07890_F00304	023	—	$f^{\prime},g^{\prime}:\mathrm{cal}\{L\}\rightarrowtail[0,1]$	$f',g':\mathcal{L}\rightarrowtail[0,1]$
2106.07890_F00305	023	—	$w_1^{\prime},w_2^{\prime\prime}\in[0,1]$	$w_1',w_2'\in[0,1]$
2106.07890_F00306	023	—	$f^{\prime}c=\exp(-f c),g^{\prime}c=\exp(-g c)$	$f'c=\exp(-f c),g'c=\exp(-g c)$
2106.07890_F00307	023	—	$w_i^{\prime\prime\prime}=\exp\left(-w_i\right)$	$w_i'=\exp(-w_i)$
2106.07890_F00308	024	—	$d_{\mathrm{cal}\{M\}}(x,-)$	$d_{\mathcal{M}}(x,-)$
2106.07890_F00309	024	—	$((-\infty,\infty],\oplus,\odot)$	$((-\infty,\infty],\oplus,\odot)$
2106.07890_F00310	024	—	$([0,\infty],\oplus,\odot)$	$([0,\infty],\oplus,\odot)$
2106.07890_F00311	024	—	$w_1=w_2=0$	$w_1=w_2=0$
2106.07890_F00312	024	—	$[0,\infty]\times\widehat{\mathrm{cal}}\{M\}\rightarrowtail\widehat{\mathrm{cal}}\{M\}$	$[0,\infty]\times\widehat{\mathcal{M}}\rightarrowtail\widehat{\mathcal{M}}$
2106.07890_F00313	024	—	$(s,f)\mapsto s\odot f$	$(s,f)\mapsto s\odot f$
2106.07890_F00314	024	—	$(s\cdot f)(x)=f(x)+s$	$(s\cdot f)(x)=f(x)+s$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
2106.07890_F00315	024	—	$\left(s_1 \oplus s_2\right) f(x)=\min \left\{s_1, s_2\right\} f(x)=\min \left\{f x+s_1, f x+s_2\right\}$	$\left(s_1 \oplus s_2\right) f(x)=\min \left\{s_1, s_2\right\} f(x)=\min \left\{f x+s_1, f x+s_2\right\}$
2106.07890_F00316	024	—	$\left(s_1 \odot s_2\right) f(x)=f x+s_1+s_2$	$\left(s_1 \odot s_2\right) f(x)=f x+s_1+s_2$
2106.07890_F00317	024	—	$\left(s_1\right) \odot\left(s_2 \odot f\right)(x)=$	$\left(s_1\right) \odot\left(s_2 \odot f\right)(x)=$
2106.07890_F00318	024	—	$\left(s_2 \odot f\right)(x)+s_1=f x+s_2+s_1$	$\left(s_2 \odot f\right)(x)+s_1=f x+s_2+s_1$
2106.07890_F00319	026	—	$\sqcup$	$\sqcup$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Unrecovered image regions — TikZ / table / failed-math candidates

Regions MathPix left as images (no LaTeX). Reconstruct with pdfdrill vision (LLM classify + transcribe to TikZ/tabular/math); verify an LLM result against the real ink with inkdrill.

Identifier	Page	Image	Source
2106.07890_DIA_0001	005		tex.zip (local)
2106.07890_DIA_0002	009		tex.zip (local)
2106.07890_DIA_0003	012		tex.zip (local)
2106.07890_DIA_0004	016		tex.zip (local)
2106.07890_DIA_0005	016		tex.zip (local)
2106.07890_DIA_0006	022		tex.zip (local)